

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

What's All This Analog IC Design Stuff, Anyhow?

How to Think Like an Analog IC Designer

Christoph Maier

Hardware, Software, & Elsewhere

Integrated Systems Neuroengineering Laboratory
University of California San Diego

11 March 2019



Table of Contents

Why?

Problem

CMOS nonidealities

Solution

Design Patterns

Matching design pattern

Resistors

Capacitors

Transistors

Differential design pattern

Current mirror

Differential pair

Fully differential design

More examples

Beyond patterns

Questions

1 Why do analog IC design?

2 Problems with IC technology

3 Solution

4 Matching design pattern

5 Differential design pattern

6 Analog Design Thought that Doesn't Fit a Pattern

7 Questions



Motivation

Analog IC Design Patterns

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

I've been told that if you're interviewing for a faculty position, you're supposed to show some *"motivate the audience"* slides





Why do analog IC design?

Like,

Generic Sales Talk

MY LATEST DEEP-LEARNING COGNITIVE SMART ULTRA-WIDEBAND ENERGY-HARVESTING AUTONOMOUS $GADGET WITH INTEGRATED BRAIN-COMPUTER INTERFACE MAY REVOLUTIONIZE INTERNET-OF-THINGS



I Hate Sales Talk

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

. . . but

I HATE SALES TALK!!!

and I'm really, **really** bad at it.

So I need your help with this.



Self-Driven Audience Motivation [1]

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

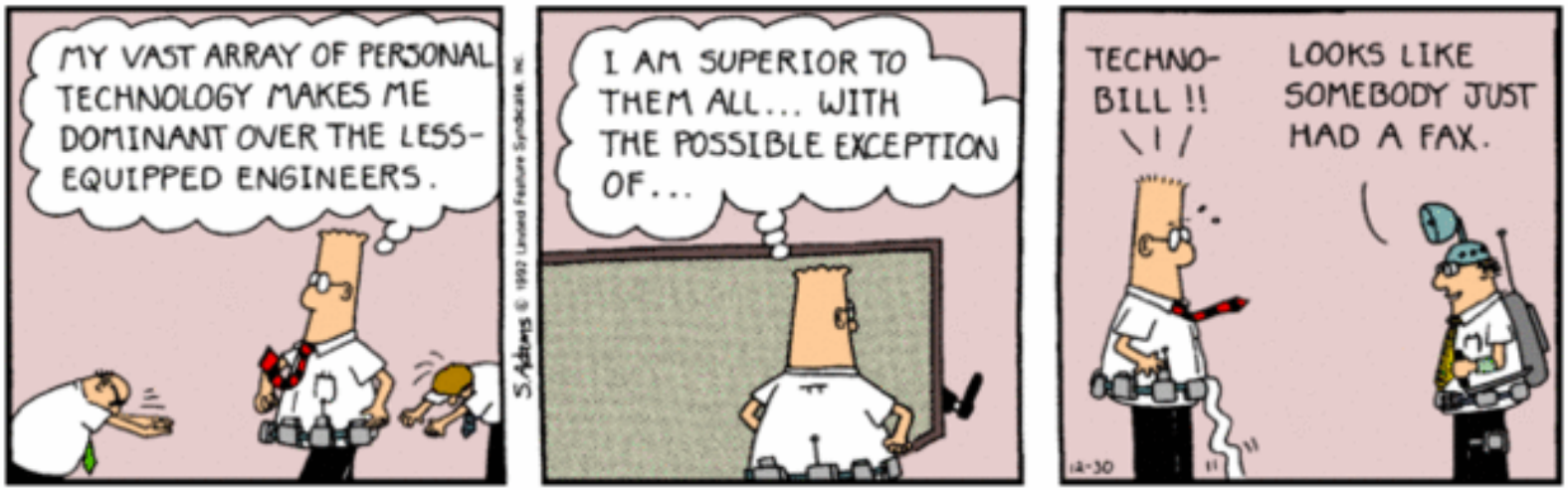
Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

Some kind of *"show of hands"* motivational trick is old and *boring*.
Let's try a **"show of gadgets"** instead.





Self-Driven Audience Motivation [2]

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

Who carries a smart phone?

Does it

- have a camera? *CMOS imager*
- know which way is up? *MEMS accelerometer*
- have a magnetic compass? *integrated magnetometer*
- have GPS? *GPS RF receiver*

Anyone wrote their own app for the sensors in their phone?



Self-Driven Audience Motivation [3]

(don't share information you're not comfortable sharing)

Any Cyborgs in the room?

Who wears

- a fitness tracker? *accelerometer, pulse oxymeter*
- a hearing aid? *integrated microphone, audio amplifier and filter*
- a glucose meter? *chemical sensor*
- maybe even an implanted device? *ECG sensor, stimulator, defibrillator*



Are You Motivated Yet?

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

I hope that you have done a good enough job to motivate yourselves that

- accurate
- precise
- robust

analog IC design might be useful and important.



Table of Contents

Why?

Problem

CMOS nonidealities

Solution

Design Patterns

Matching design pattern

Resistors

Capacitors

Transistors

Differential design pattern

Current mirror

Differential pair

Fully differential design

More examples

Beyond patterns

Questions

1 Why do analog IC design?

2 Problems with IC technology
- CMOS process nonidealities

3 Solution

4 Matching design pattern

5 Differential design pattern

6 Analog Design Thought that Doesn't Fit a Pattern

7 Questions



Problems peculiar to analog CMOS IC design

Why?

Problem

CMOS nonidealities

Solution

Design Patterns

Matching design pattern

Resistors

Capacitors

Transistors

Differential design pattern

Current mirror

Differential pair

Fully differential design

More examples

Beyond patterns

Questions

Problems analog IC designers face all the time:

Component parameters vary substantially (typically $\pm 50\%$)

- between individual dice
- over temperature
- Redeeming quality: Adjacent devices on same die match well (typ. $\pm 1\%$)

Integrated microsensors

- have fabrication tolerances
- have cross-sensitivities



CMOS processes: Designed for digital logic

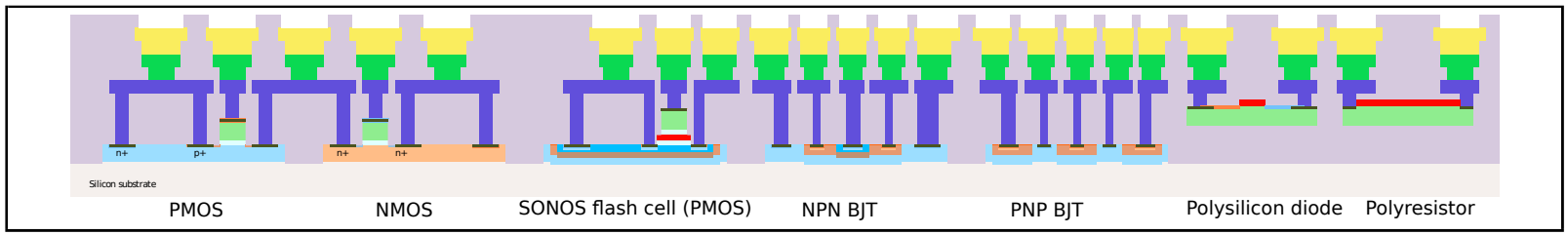
- Digital logic: optimized for maximum insensitivity to circuit device properties
  - resistance of intentional and parasitic resistors
  - capacitance of intentional and parasitic capacitors
  - transconductance g_m and threshold voltage V_T of MOSFETs
- CMOS feature size designed accordingly
- Rough ballpark figures:
  - $\frac{\Delta R}{R} \approx 50\%$
  - $\frac{\Delta C}{C} \approx 30\%$
  - $\frac{\Delta g_m}{g_m} \approx 30\%$
  - $\frac{\Delta V_T}{V_T} \approx 50\%$



CMOS processes: Different devices vary differently

Different active and passive devices available in IC process

- Transistors: NMOS, PMOS, ...
- Resistors: well, diffusion, polysilicon
- Capacitors: gate, interconnect, poly-poly



David Lanzendörfer, *LibreSilicon process HKUST (NFF)*, December 15, 2018, presented at 35C3, Leipzig

Device properties depend on different processing steps



Large inter-die vs. small intra-die parameter variations

- Large parameter changes between different wafers, IC dice
- Variation of device parameters on one IC die is small (few %)

Example:

Threshold voltage of two transistors next to each other:

$$\begin{pmatrix} V_{T1} \\ V_{T2} \end{pmatrix} =: \begin{pmatrix} 1 & 1 & 1/2 \\ 1 & 1 & -1/2 \end{pmatrix} \cdot \begin{pmatrix} V_{T,mean} \\ \Delta V_{T,d2d} \\ \delta V_T \end{pmatrix}$$

where typical die-to-die and intra-die variabilities are, respectively

$$\Delta V_{T,d2d} \approx V_{T,mean}/2 \quad \text{and} \quad \delta V_T \approx V_{T,mean}/200$$

For less handwaving numbers, see, e. g.:

P. Kinget, "Device Mismatch and Tradeoffs in the Design of Analog Circuits", IEEE JSSC, June 2005



Table of Contents

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

1 Why do analog IC design?

2 Problems with IC technology

3 Solution
- Design patterns

4 Matching design pattern

5 Differential design pattern

6 Analog Design Thought that Doesn't Fit a Pattern

7 Questions



Similar Problems ⇒ Similar Solutions

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

Design pattern

- Typical solution to solve a recurring design problem
- General concept for solving a particular problem
- High-level description of a solution
- Same pattern applied to two different problems will be different

Source: https://refactoring.guru/design-patterns/what-is-pattern



Example: Digital Design Anti-Pattern

Digital resolution: number of significant bits
Subtract two almost equal 16-bit numbers:

$$\begin{array}{r} 1100\,1010\,1111\,1110 \\ -1100\,1010\,1111\,1010 \\ \hline 0000\,0000\,0000\,0100 \end{array}$$

Result has only 3 significant bits left
$\Rightarrow$ Loss of factor $2^{13} = 8192$ of resolution.

For precise digital numerical computations, subtracting almost equal numbers is a **design anti-pattern**.



Analog Design Patterns

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

Design patterns for precision analog design

Device Matching

- Use devices of the same type
- Use unit devices of same geometry
- Use same orientation
- Avoid edge effects

Differential Design

Use difference of signals by identical, adjacent components
In analog design, subtracting almost equal numbers **improves** precision.



Table of Contents

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

1 Why do analog IC design?

2 Problems with IC technology

3 Solution

4 Matching design pattern
- Resistors
- Capacitors
- Transistors

5 Differential design pattern

6 Analog Design Thought that Doesn't Fit a Pattern



Resistor matching

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

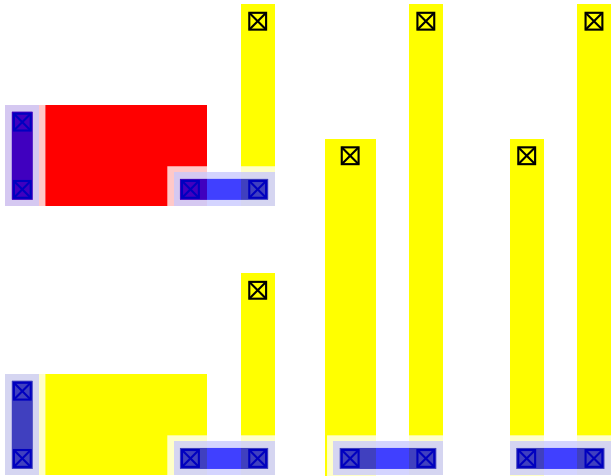
Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

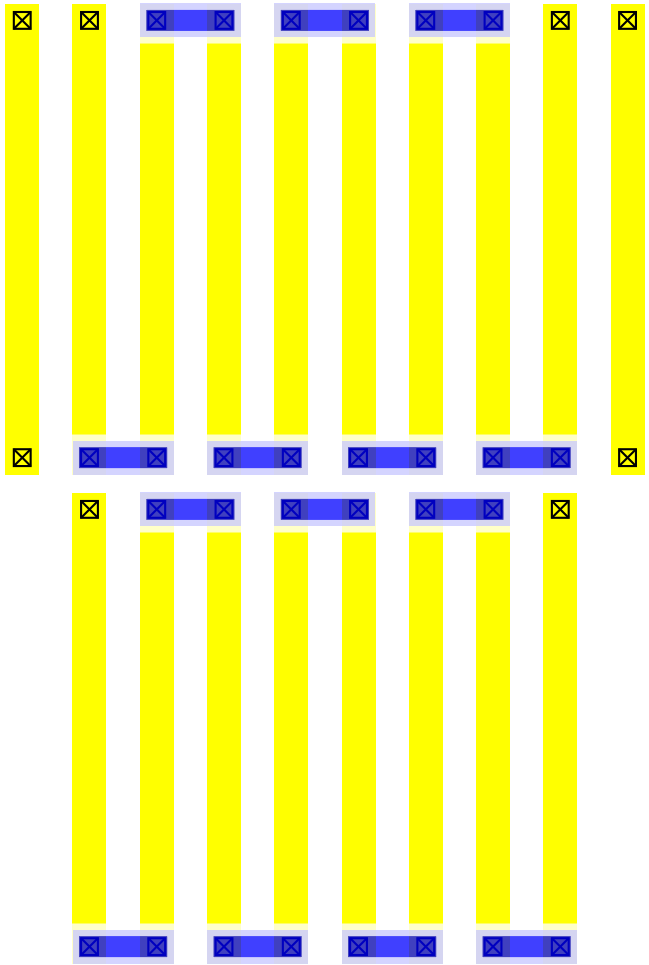
Beyond patterns

Questions

Use

- same type resistors
- same orientation
- same width
- unit devices
- dummy devices





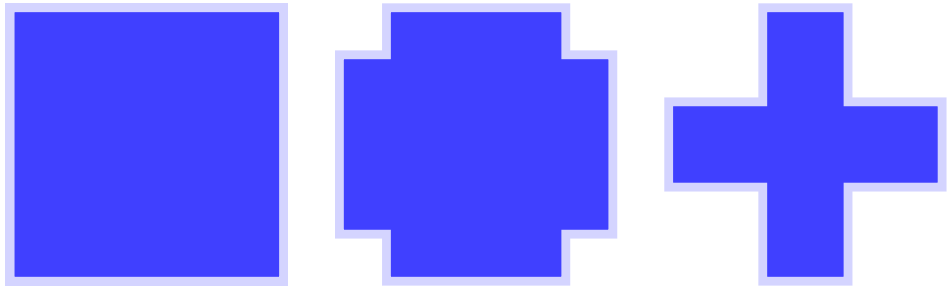


Capacitor matching

- use same type capacitors
- Two components vary independently
  - Area capacitance $C_\square$, dimension $\mathrm{fF}/\,\mu\mathrm{m}^2$
  - Perimeter capacitance C_P, dimension $\mathrm{fF}/\,\mu\mathrm{m}$
- Match $\frac{C_P}{C_\square}$ ratio
- Use unit devices
- Use dummy devices

Example: Cross structure, C_P stays the same
Length: L Width: W Cutouts: l_1, l_2, w_1, w_2
$C = C_\square \cdot L \cdot W - l \cdot w + 2\,C_P \cdot (L + W)$

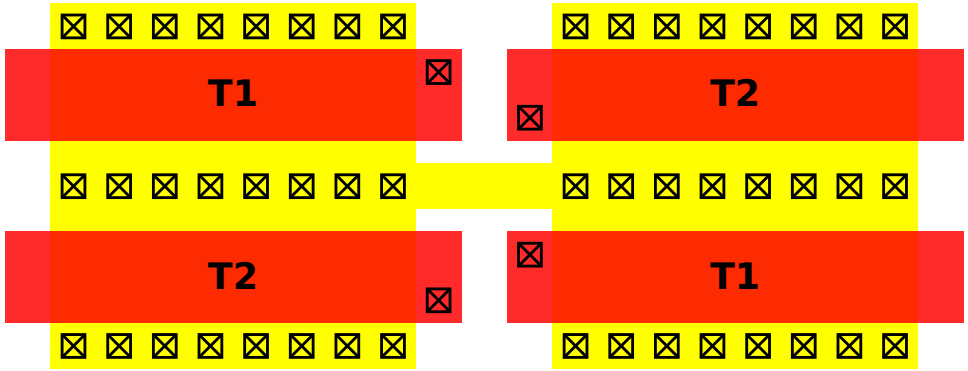
with $l := l_1 + l_2$ and $w := w_1 + w_2$





Transistor matching

- Do not use minimum feature size
- Use same type transistors
- Use unit devices
- Common centroid design



- Use dummy devices



Table of Contents

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

1 Why do analog IC design?

2 Problems with IC technology

3 Solution

4 Matching design pattern

5 Differential design pattern
- Current mirror
- Differential pair
- Fully differential design
- More examples

6 Analog Design Thought that Doesn't Fit a Pattern



Subtracting large quantities: Current mirror

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Why?

Problem
CMOS nonidealities

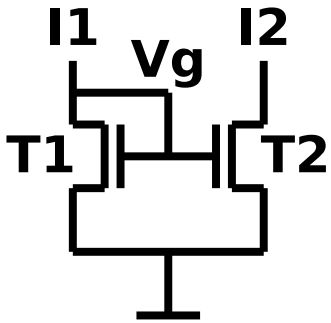
Solution
Design Patterns

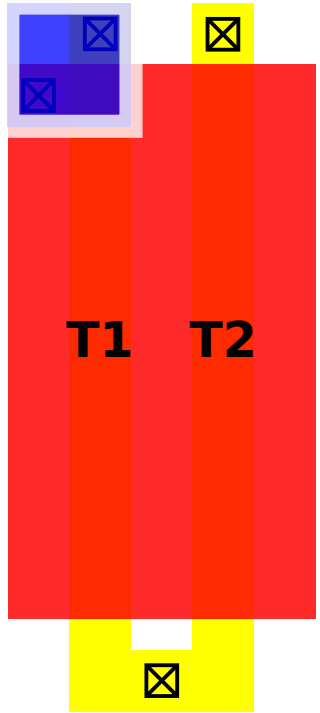
Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions







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Analog IC Design Patterns

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Why?

Problem
CMOS nonidealities

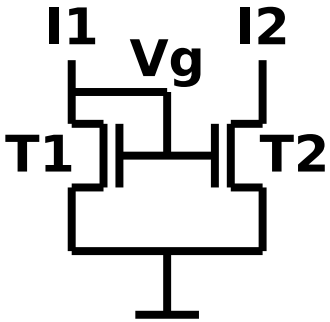
Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions



Transistor current: $I_d =: G[V_g - V_T]$

Common gate voltage: $V_g = G^{-1}[I_1] + V_{T1}$

Output current:

$$I_2 = G[V_g - V_{T2}] = G\left[G^{-1}[I_1] + V_{T1} - V_{T2}\right]$$

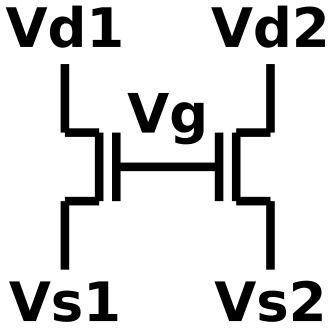
$$V_{T1} - V_{T2} = \underbrace{\begin{pmatrix} 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 1/2 \\ 1 & 1 & -1/2 \end{pmatrix}}_{\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}} \cdot \begin{pmatrix} V_{T,mean} \\ \Delta V_{T,d2d} \\ \delta V_T \end{pmatrix} = \delta V_T$$

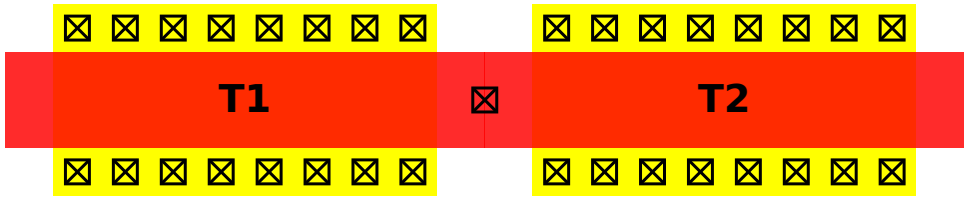
$$I_2 \approx I_1 + \left.\frac{\mathrm{d}G}{\mathrm{d}V}\right|_{I_1} (V_{T1} - V_{T2}) =: I_1 + g_m \delta V_T$$



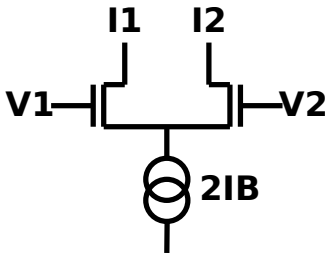
Subtracting large quantities: Differential pair

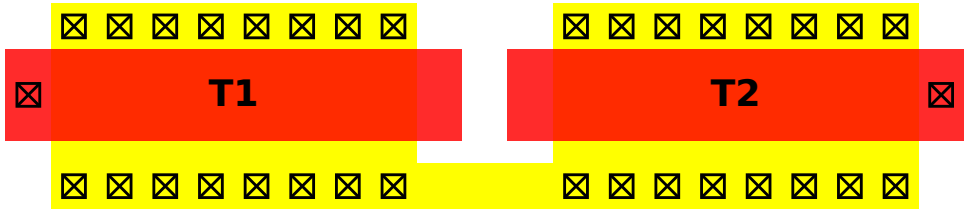
- Common base differential pair





- Common source differential pair







Subtracting large quantities: Differential pair

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Why?

Problem
CMOS nonidealities

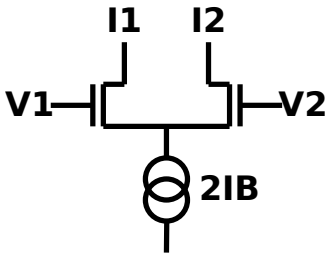
Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions



Transistor current:

$$I_d =: G[V_g - V_s - V_T]$$

Split V, I into differential and common mode signals

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} 1/2 & 1 \\ -1/2 & 1 \end{pmatrix} \cdot \begin{pmatrix} \Delta V \\ V_g \end{pmatrix}, \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} 1/2 & 1 \\ -1/2 & 1 \end{pmatrix} \cdot \begin{pmatrix} \Delta I \\ I_B \end{pmatrix}.$$

Common source voltage for both transistors:

$$\begin{pmatrix} V_s \\ V_s \end{pmatrix} = \begin{pmatrix} V_1 - V_{T1} - G^{-1}[I_1] \\ V_2 - V_{T2} - G^{-1}[I_2] \end{pmatrix}$$

$$0 = \underbrace{\begin{pmatrix} 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1/2 & 1 & -1 & -1 & -1/2 & -1 & 0 \\ -1/2 & 1 & -1 & -1 & 1/2 & 0 & -1 \end{pmatrix}}_{\begin{pmatrix} 1 & 0 & 0 & 0 & -1 & -1 & 1 \end{pmatrix}} \cdot \begin{pmatrix} \Delta V \\ V_g \\ V_{T,mean} \\ \Delta V_{T,d2d} \\ \delta V_T \\ G^{-1}[I_1] \\ G^{-1}[I_2] \end{pmatrix}$$

$$\Delta V - \delta V_T = \begin{pmatrix} 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} G^{-1}\left[\frac{\Delta I}{2} + I_B\right] \\ G^{-1}\left[-\frac{\Delta I}{2} + I_B\right] \end{pmatrix} \approx \underbrace{\begin{pmatrix} 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2g_m} & 1 \\ \frac{-1}{2g_m} & 1 \end{pmatrix}}_{\begin{pmatrix} \frac{1}{g_m} & 0 \end{pmatrix}} \cdot \begin{pmatrix} \Delta I & G^{-1}[I_B] \end{pmatrix}$$

$$\Delta I \approx g_m \Big|_{I_B} (\Delta V - \delta V_T)$$



Fully differential OTA

Why?

Problem
CMOS nonidealities

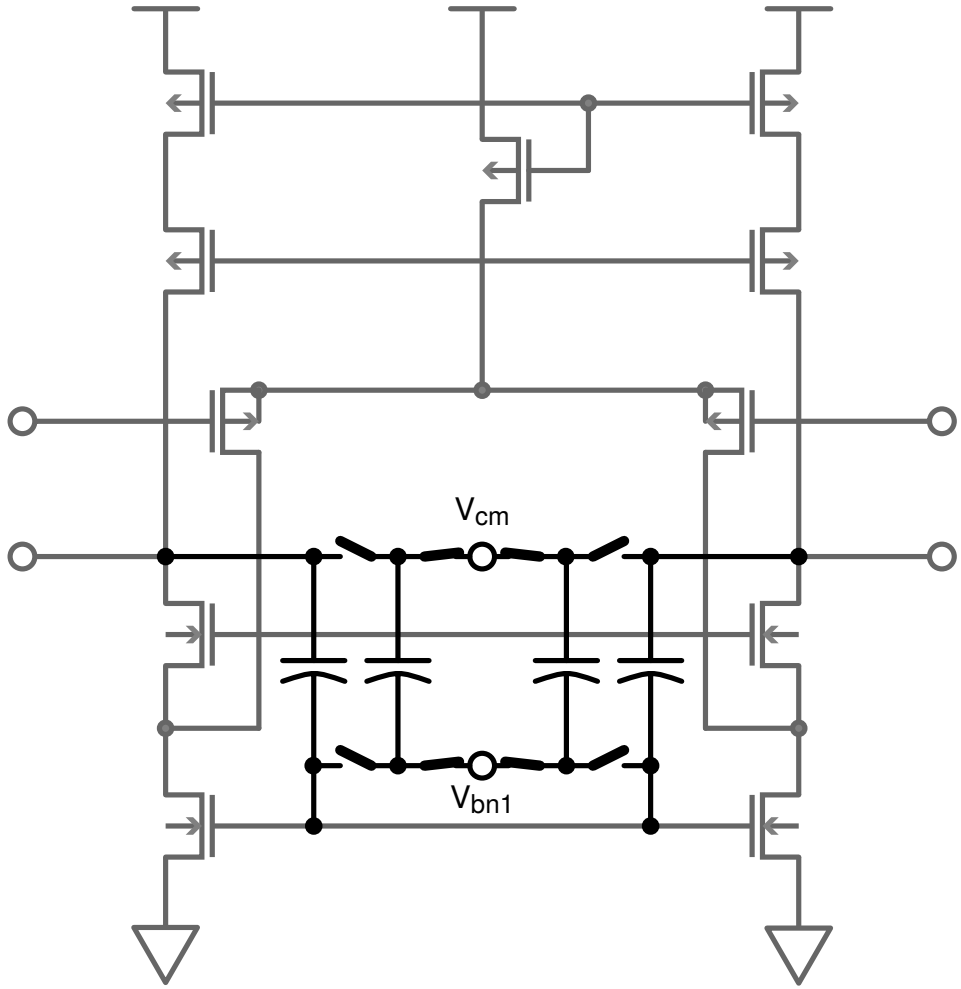
Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions





Subtracting input capacitance

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Why?

Problem
CMOS nonidealities

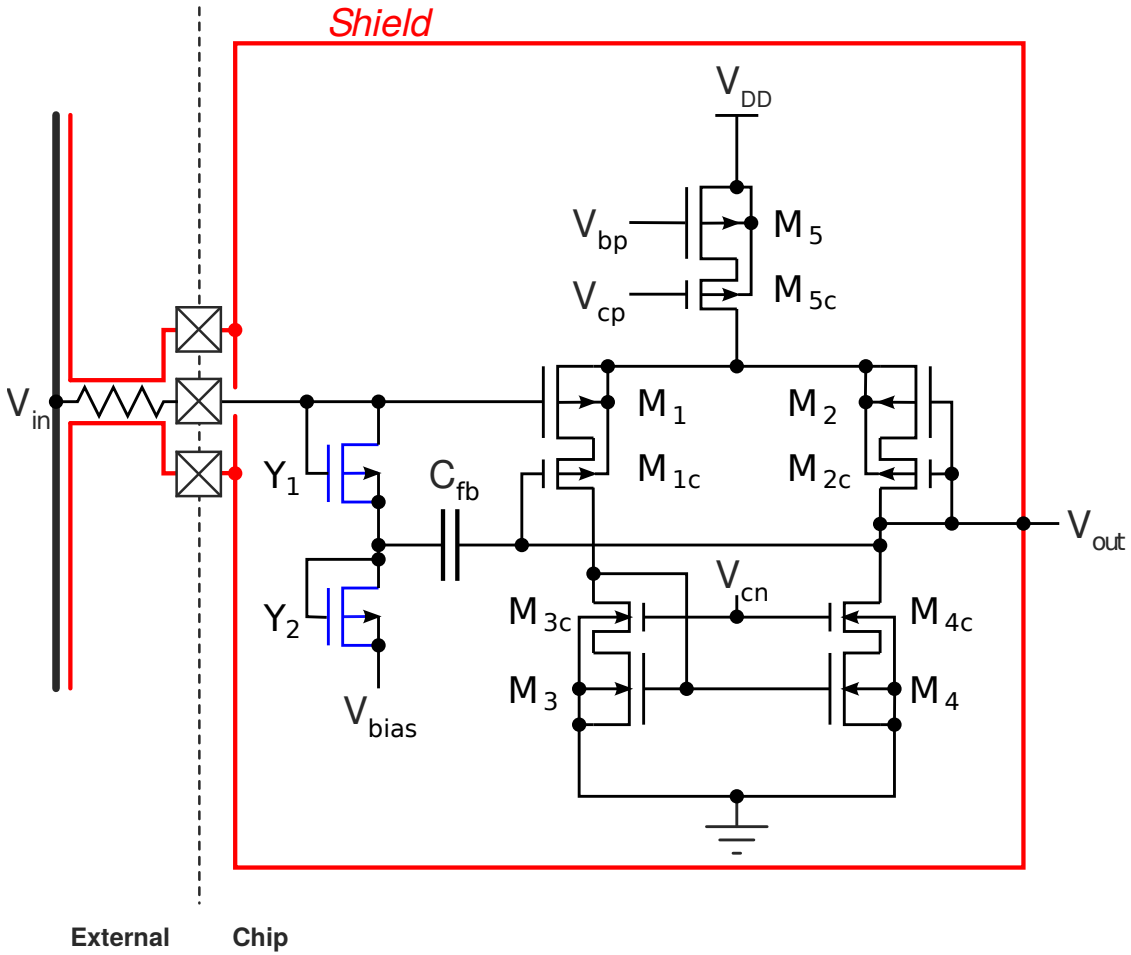
Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions





Subtracting magnetic field from mechanical stress

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

Spinning current Hall sensor



Subtracting asymmetrical signal from symmetrical signal

Analog IC Design Patterns

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

Differential fluxgate sensor



Subtraction by polarity reversal: Chopping

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions



Subtraction in time: Correlated Double Sampling

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions



Table of Contents

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

1 Why do analog IC design?

2 Problems with IC technology

3 Solution

4 Matching design pattern

5 Differential design pattern

6 Analog Design Thought that Doesn't Fit a Pattern

7 Questions



Out Of The Box

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

- For recurring Problems, Design Patterns help
- No Problems are the same in Analog Design
- Need to view Problems from Different Angles
- For Analog Design, **Think in Analogies**
- Need **deep understanding** of **both**
  - **essence**
  - **context**

  of the Problem



Analog Design Beyond Patterns

Design Patterns

Guidelines Only

Also required:

- Creative implementation
- Interdisciplinary thinking in analogies
- Independent, out-of-the box thought

Can only be taught in examples



Out-of-the-Box Example

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Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

I'll try to give an example.

This is my first lecture, ever.

So I will either crash and burn, or succeed with flying colors.



Interdisciplinary design pattern: Matching

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Why?

Problem

CMOS nonidealities

Solution

Design Patterns

Matching design pattern

Resistors

Capacitors

Transistors

Differential design pattern

Current mirror

Differential pair

Fully differential design

More examples

Beyond patterns

Questions





Interdisciplinary design pattern: Matching

Analog IC Design Patterns

Christoph Maier

Why?

Problem

CMOS nonidealities

Solution

Design Patterns

Matching design pattern

Resistors

Capacitors

Transistors

Differential design pattern

Current mirror

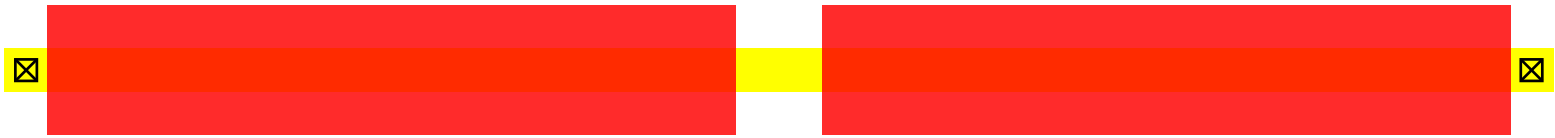
Differential pair

Fully differential design

More examples

Beyond patterns

Questions





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Analog IC Design Patterns

Christoph Maier

Why?

Problem
- CMOS nonidealities

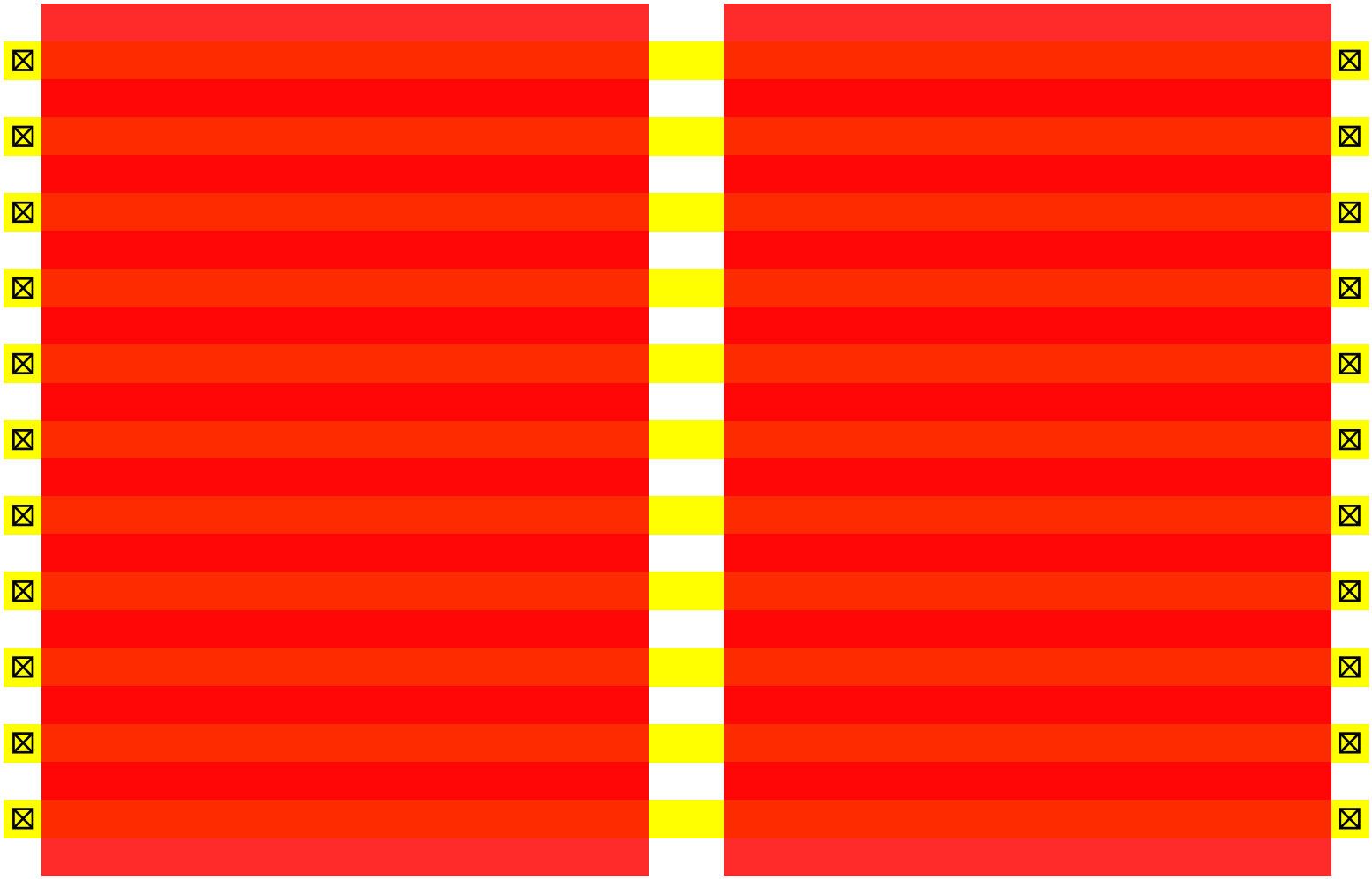
Solution
- Design Patterns

Matching design pattern
- Resistors
- Capacitors
- Transistors

Differential design pattern
- Current mirror
- Differential pair
- Fully differential design
- More examples

Beyond patterns

Questions





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Analog IC Design Patterns

Christoph Maier

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Problem
- CMOS nonidealities

Solution
- Design Patterns

Matching design pattern
- Resistors
- Capacitors
- Transistors

Differential design pattern
- Current mirror
- Differential pair
- Fully differential design
- More examples

Beyond patterns

Questions





Interdisciplinary design pattern: Matching

Analog IC Design Patterns

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- Current mirror
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Beyond patterns

Questions





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Problem
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- Resistors
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Beyond patterns

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Problem
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- Resistors
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Differential design pattern
- Current mirror
- Differential pair
- Fully differential design
- More examples

Beyond patterns

Questions





Table of Contents

Why?

Problem
CMOS nonidealities

Solution
Design Patterns

Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions

1 Why do analog IC design?

2 Problems with IC technology

3 Solution

4 Matching design pattern

5 Differential design pattern

6 Analog Design Thought that Doesn't Fit a Pattern

7 Questions



Why?

Problem
CMOS nonidealities

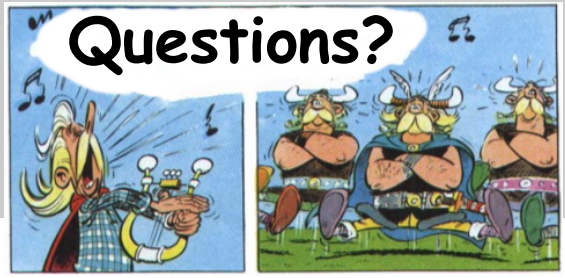
Solution
Design Patterns

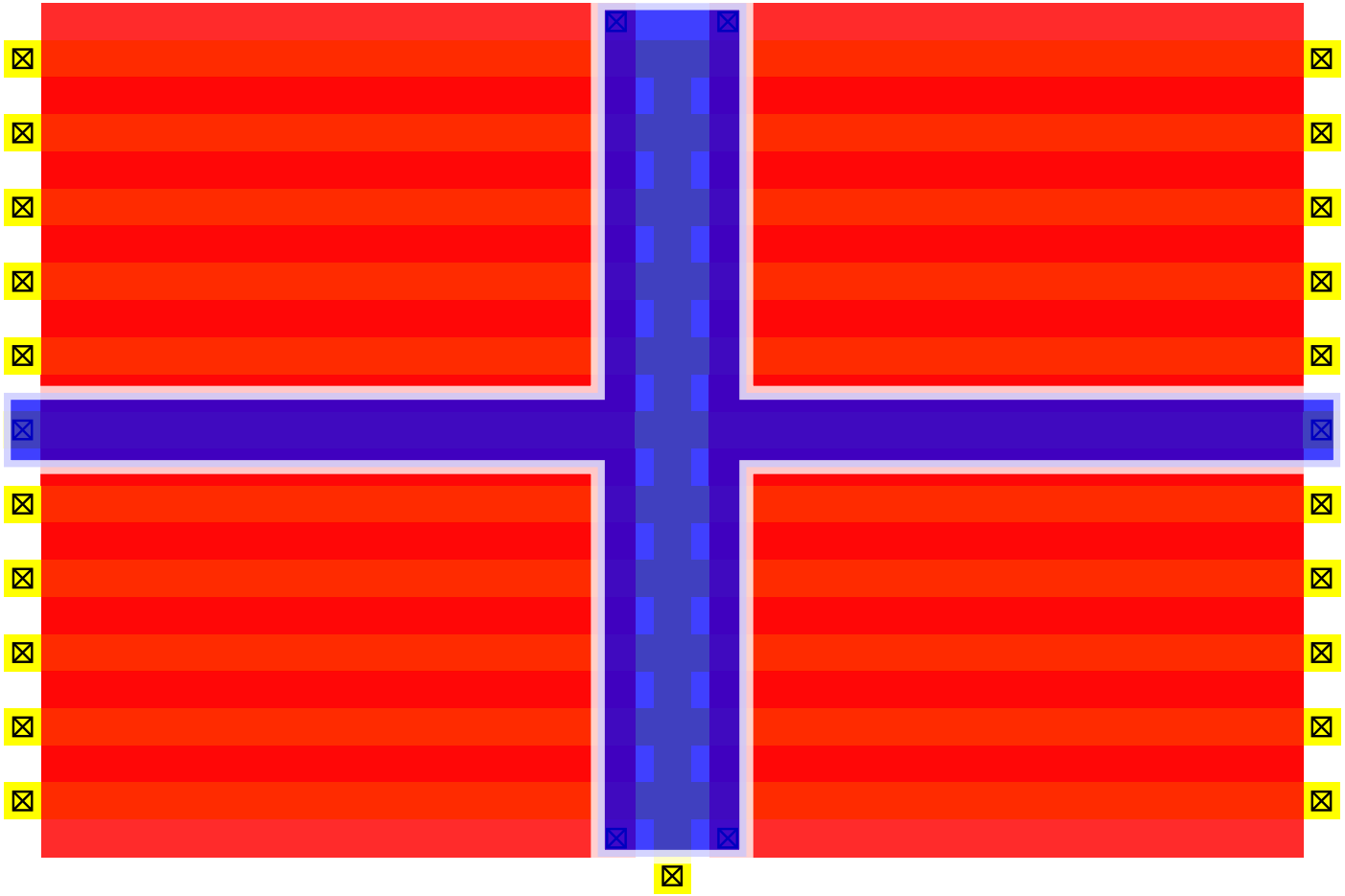
Matching design pattern
Resistors
Capacitors
Transistors

Differential design pattern
Current mirror
Differential pair
Fully differential design
More examples

Beyond patterns

Questions





https://bitbucket.org/cmucsd/analogdesignlecture/downloads/



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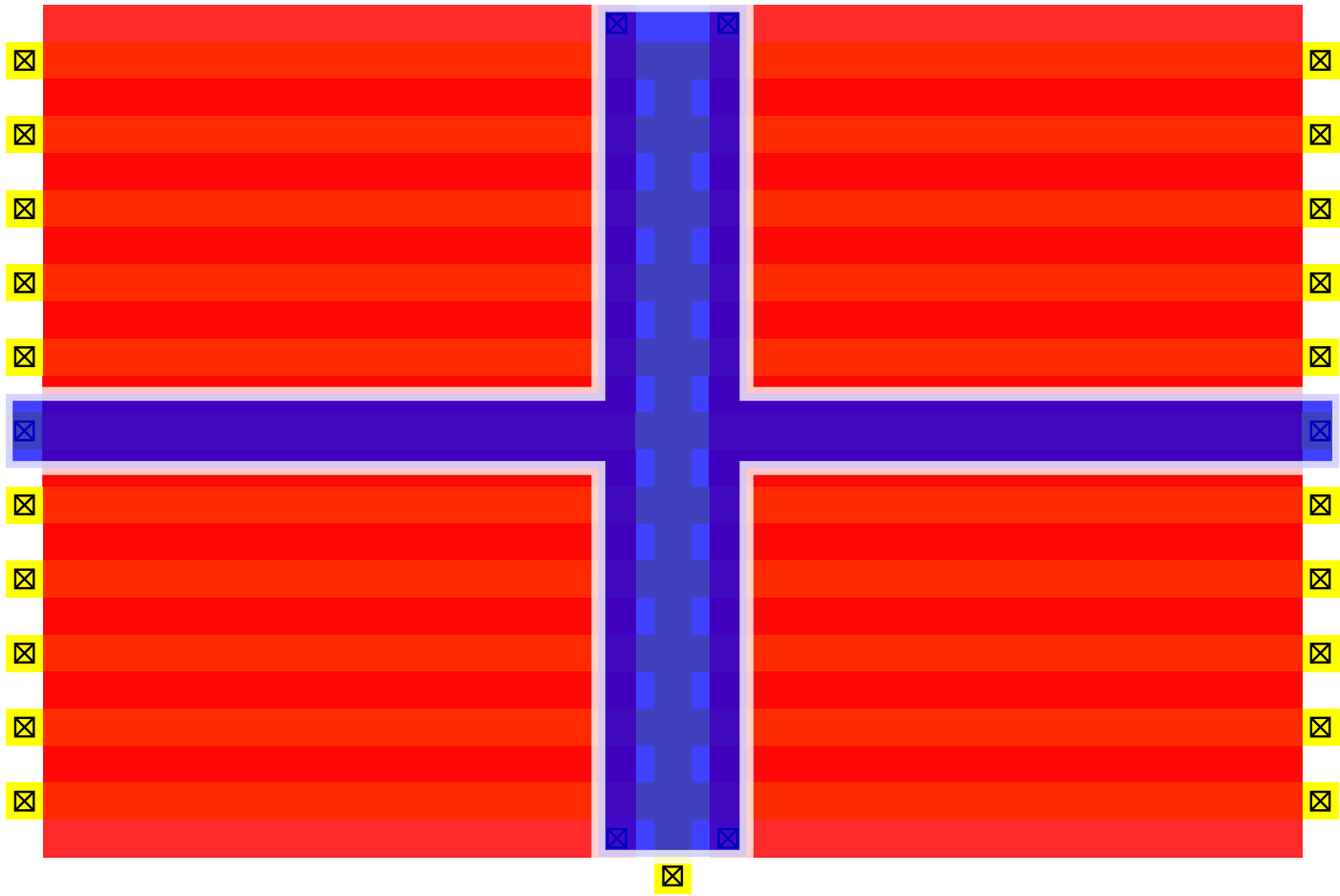
Solution
- Design Patterns

Matching design pattern
- Resistors
- Capacitors
- Transistors

Differential design pattern
- Current mirror
- Differential pair
- Fully differential design
- More examples

Beyond patterns

Questions



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